Ahmadyar Qureshi

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Engineering and Industry Experience

National Corrosion and Materials Reliability Lab (NCMRL), ***Bryan, Texas*** **January 2024 – PRESENT**

- *Graduate Research Assistant - Corrosion & Materials Integrity*, **Texas A&M University**, College Station, Texas
- Evaluated corrosion and materials performance of sucker-rod steels under HPHT sour-service environments containing **100% CO_2, 100% H_2S, and 50/50 H_2S/CO_2**, using an integrated high-pressure testing and analysis setup
- Investigated hydrogen embrittlement susceptibility and residual-stress effects in additively manufactured 316LN stainless steel following electrochemical hydrogen charging (**50 mA/cm², 3 days**), correlating electrochemical measurements with material degradation
- Evaluated **additively manufactured Cu-Ni alloys** for corrosion and **stress-corrosion cracking (SCC)** susceptibility, examining the influence of **2 build orientations and temperatures up to 80 °C** in aggressive chloride environments on material integrity
- Tested carbon- and glass-fiber-reinforced polymer composites under pressurized hydrogen at **35 and 70 bar total pressure for 3–5 days** to determine degradation and suitability for hydrogen storage and transportation infrastructure
- Engineered Co-, Cu-, and CoCu-based electrocatalysts with telluride modifications for **hydrogen production via water electrolysis**, assessing **HER performance** through LSV, CV, Tafel analysis, EIS, and durability testing in **3 alkaline electrolyte environments**
- Conducted **mechanical and fracture testing** of metallic materials, including tensile, SSRT, hardness, CTOD/CMOD, and hydrogen embrittlement testing, to characterize mechanical integrity, cracking susceptibility, and failure behavior under service-relevant conditions
- **Executed proprietary industry-sponsored corrosion testing of α-HESA acid systems**, including electrochemical/materials performance assessment under application-relevant conditions
- Coordinated specimen fabrication and preparation via **EDM, waterjet cutting, machining, and controlled thermomechanical processing** for mechanical, corrosion, and materials qualification testing

SET Laboratories Inc., ***Houston, Texas*** **May 2026 – Present**

- *Corrosion Scientist Intern – CCUS & Pipeline Integrity*
- Assessed **API 5L X52/X65 pipeline steels** under dense and supercritical CO_2 environments to quantify corrosion risks associated with CCUS transportation
- Designed and executed **50 °C/80 bar** autoclave testing under relevant H_2O and impurity conditions representative of CO_2 transportation streams
- Examined effects of **H_2O, O_2, SO_2, NO_2, and H_2S** impurities on uniform and localized corrosion behavior
- Developed experimental protocols for impurity-controlled dense/supercritical CO_2 testing, including fluid preparation, transfer, exposure, post-test characterization, and corrosion-rate assessment
- Generated materials-performance data supporting an **industry consortium** focused on pipeline integrity and CO_2 transportation

Center for Advanced Materials, ***Doha, Qatar*** **September 2021 – October 2023**

- *Undergraduate Research Assistant*, **Qatar University**
- Produced and validated **smart corrosion-protective coating systems** incorporating inhibitor-loaded mesoporous ceria and talc carriers into polymeric matrices for metallic substrate protection
- Formulated **pH-responsive inhibitor systems** using sodium benzoate and 8-hydroxyquinoline to provide active corrosion protection and structured inhibitor release
- Performed coating/material characterization using **EIS, salt-spray exposure, SEM/TEM, XRD, FTIR, and UV–Vis** to correlate coating chemistry, morphology, and barrier performance
- Optimized particle synthesis, inhibitor loading, and coating formulation, achieving approximately **10^9 Ω·cm² impedance** and substantially improved long-term corrosion resistance

Chiyoda Al Mana engineering LLC., ***Doha, Qatar*** **June 2023 – August 2023**

- Supported **brownfield LNG and wastewater-reuse engineering projects**, reviewing process modifications and existing facility constraints
- Interpreted and reviewed **PFDs and P&IDs** to understand equipment, piping, process streams, and proposed modifications
- Completed **hydraulic and depressurization calculations** supporting process engineering evaluations
- Prepared engineering documentation and process workflows supporting brownfield modifications and project deliverables

Education

PhD in Materials Science and Engineering (MSEN), Texas A&M University, College Station, Texas **May 2026 – Present**

- GPA: 3.60/4.0 | Expected graduation December 2027

MSc in Materials Science and Engineering (MSEN), Texas A&M University, College Station, Texas **January 2024 – May 2026**

- GPA: 3.60/4.00 | **Best MS Thesis Award**

BSc in Chemistry, Qatar University, Doha, Qatar **January 2020 – December 2023**

- GPA: 3.40/4.00 | Dean's List (1 year) | Vice-President's List (2 years)

Technical Expertise

Corrosion & Integrity: HPHT Corrosion | Sour Service | CO_2/H_2S Corrosion | CCUS Pipeline Integrity | Hydrogen Embrittlement | SCC | Localized Corrosion | Protective Coatings
Electrochemistry: EIS | LPR | CPP | OCP/Ecorr | Potentiodynamic Polarization | Hydrogen Charging
HPHT Testing: Autoclave Systems | Dense/Supercritical CO_2 | Impurity-Controlled Testing | CO_2/H_2S Environments
Hydrogen & Energy: Hydrogen Materials | Water Electrolysis/HER | Electrocatalysis | Hydrogen Storage & Transportation
Materials & Characterization: Additive Manufacturing | SEM/EDS | EBSD | XRD | TDS | TEM | FTIR | Profilometry
Engineering: PFD/P&ID Interpretation | Hydraulic Calculations | Depressurization | Experimental/Test Protocol Development
Leadership: Vice President, American Chemical Society – Qatar University Chapter | Led student teams, technical outreach, and international collaborations